Skills Summary

Differentiating Implicit Relations

Use this reference while completing your worksheet or when studying.

Skill 1 — The Basic Move

Rule: differentiate both sides with respect to x. A term in x behaves normally; a term in y picks up a chain-rule factor:

$$\frac{d}{dx}\left(y^n\right) = ny^{n-1}\frac{dy}{dx}.$$

Then gather the $\frac{dy}{dx}$ terms on one side, factor it out, and divide. The answer normally contains both x and y, and that is correct — a relation can have two different slopes above the same x.

Example: For $x^2 + 2y^2 = 22$, find the slope at $(2, 3)$.
Step 1. The relation cannot be solved for y in one clean branch here, so differentiate as it stands and let the chain rule carry the dy/dx.
Step 2. $2x + 4y\frac{dy}{dx} = 0$ gives $\frac{dy}{dx} = -\frac{x}{2y}$, and at $(2, 3)$ that is $-\frac{2}{6}$.
$\Rightarrow$ $\mathbf{-\frac{1}{3}}$

Try it: For $4x^2 + y^2 = 20$, find the slope at $(1, 4)$.

check: -1

Skill 2 — Products and Powers of y

Rule: a term like xy or x^2y^3 is a *product* of a function of x and a function of x — because y is one too. So the product rule applies, and the y factor still contributes its chain-rule $\frac{dy}{dx}$:

$$\frac{d}{dx}(xy) = y + x\frac{dy}{dx}.$$

Missing the first term is the single most common error, and it usually leaves a suspiciously simple answer.

Example: For $xy + y^2 = 3$, find the slope at $(2, 1)$.
Step 1. The xy term is a product, so it needs both pieces of the product rule before anything can be collected.
Step 2. $y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$ gives $\frac{dy}{dx} = -\frac{y}{x+2y}$, and at $(2, 1)$ that is $-\frac{1}{4}$.
$\Rightarrow$ $\mathbf{-\frac{1}{4}}$

Try it: For $3xy + y^2 = 18$, find the slope at $(1, 3)$.

check: -1

Skill 3 — Checking Against the Explicit Branch

Why it works: when a relation *can* be solved for y near a point, the implicit answer and the explicit answer must agree there — they describe the same tangent line. Substituting y back into the implicit form is the fastest way to confirm you did not lose a factor, and it is a good habit on any circle, parabola or ellipse.

Example: The circle $x^2 + y^2 = 169$ contains $(5, 12)$. Compare the two methods there.
Step 1. Near $(5, 12)$ the upper branch is $y = \sqrt{169 - x^2}$, so both routes are available and should give the same number.
Step 2. Explicitly, $\frac{dy}{dx} = \frac{-x}{\sqrt{169-x^2}}$; implicitly, $2x + 2y\frac{dy}{dx} = 0$ gives $\frac{dy}{dx} = -\frac{x}{y}$, and at $(5, 12)$ both read $-\frac{5}{12}$.

$$\Rightarrow \quad \boldsymbol{\frac{-x}{\sqrt{169 - x^2}}} \quad \text{and at the point} \quad \boldsymbol{-\frac{5}{12}}$$

Try it: The circle $x^2 + y^2 = 100$ contains $(6, 8)$. Find the explicit derivative of the upper branch, and the implicit slope at that point.

check: $\boldsymbol{\frac{-x}{\sqrt{100-x^2}}}$ and $\boldsymbol{-\frac{3}{4}}$